

ECO387C - Fall 2024

Problem Set 6 Solutions

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Question 1

The Lagrangian corresponding to the agent's maximization problem is the following:

$$\begin{aligned}\mathcal{L} = \mathbb{E}_t \bigg\{ \sum_{j=0}^{\infty} \beta^j & \left[\frac{c_{t+j}^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} + \lambda_{t+j} \left(b_{t+j-1}(1+r_{t+j-1}) + (q_{t+j} + d_{t+j})s_{t+j-1} \right. \right. \\ & + (q_{t+j} + d_{t+j} - \bar{p}^c) \mathbb{1} \{q_{t+j} + d_{t+j} - \bar{p}^c \geq 0\} o_{t+j-1}^c \\ & + (\bar{p}^p - q_{t+j} - d_{t+j}) \mathbb{1} \{\bar{p}^p - q_{t+j} - d_{t+j} \geq 0\} o_{t+j-1}^p \\ & \left. \left. - c_{t+j} - b_{t+j} - q_{t+j}s_{t+j} - p_{t+j}^c o_{t+j}^c - p_{t+j}^p o_{t+j}^p \right) \right] \bigg\}\end{aligned}$$

The first-order conditions are the following:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c_t} = 0 & \implies c_t^{-\frac{1}{\sigma}} = \lambda_t \\ \frac{\partial \mathcal{L}}{\partial b_t} = 0 & \implies \lambda_t = \beta \mathbb{E}_t \lambda_{t+1} (1 + r_t) \\ \frac{\partial \mathcal{L}}{\partial s_t} = 0 & \implies \lambda_t q_t = \beta \mathbb{E}_t \lambda_{t+1} (q_{t+1} + d_{t+1}) \\ \frac{\partial \mathcal{L}}{\partial o_t^c} = 0 & \implies \lambda_t p_t^c = \beta \mathbb{E}_t \lambda_{t+1} (q_{t+1} + d_{t+1} - \bar{p}^c) \mathbb{1} \{q_{t+1} + d_{t+1} - \bar{p}^c \geq 0\} \\ \frac{\partial \mathcal{L}}{\partial o_t^p} = 0 & \implies \lambda_t p_t^p = \beta \mathbb{E}_t \lambda_{t+1} (\bar{p}^p - q_{t+1} - d_{t+1}) \mathbb{1} \{\bar{p}^p - q_{t+1} - d_{t+1} \geq 0\}\end{aligned}$$

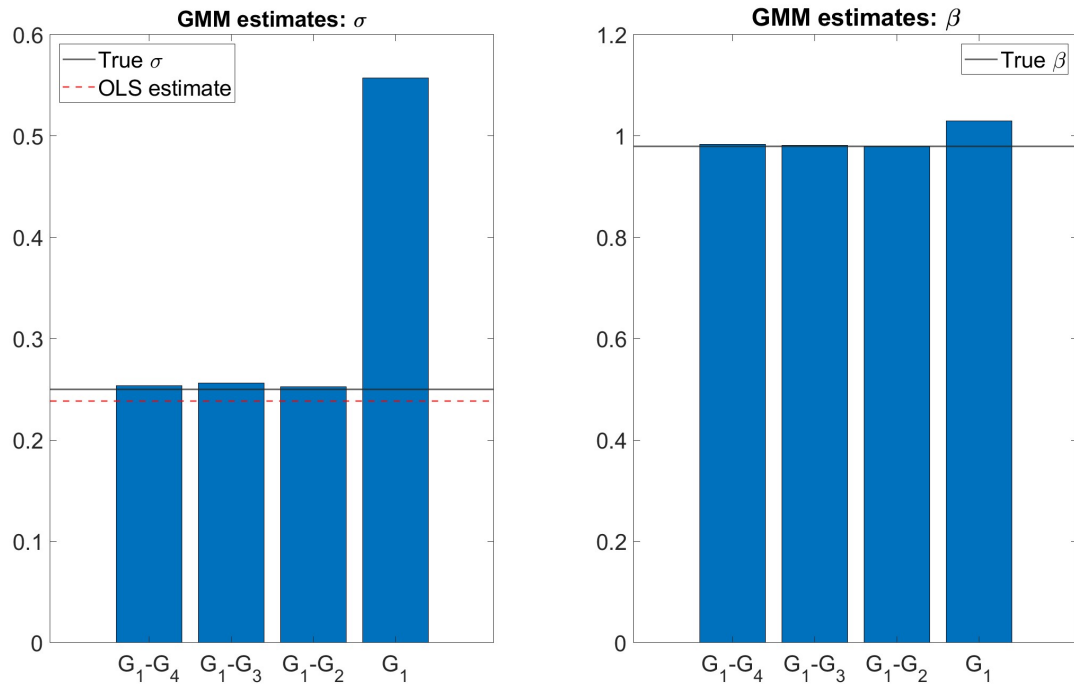
Combining these equations gives us

$$\begin{aligned}1 &= \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\frac{1}{\sigma}} (1 + r_t) \right] \\ q_t &= \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\frac{1}{\sigma}} (q_{t+1} + d_{t+1}) \right] \\ p_t^c &= \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\frac{1}{\sigma}} (q_{t+1} + d_{t+1} - \bar{p}^c) \mathbb{1} \{q_{t+1} + d_{t+1} - \bar{p}^c \geq 0\} \right] \\ p_t^p &= \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\frac{1}{\sigma}} (\bar{p}^p - q_{t+1} - d_{t+1}) \mathbb{1} \{\bar{p}^p - q_{t+1} - d_{t+1} \geq 0\} \right]\end{aligned}$$

Rearranging the terms gives us the desired form:

$$\begin{aligned}
0 &= \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\frac{1}{\sigma}} (1 + r_t) - 1 \right] \\
0 &= \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\frac{1}{\sigma}} \frac{(q_{t+1} + d_{t+1})}{q_t} \right] \\
0 &= \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\frac{1}{\sigma}} \frac{(q_{t+1} + d_{t+1} - \bar{p}^c)}{p_t^c} \mathbb{I} \{q_{t+1} + d_{t+1} - \bar{p}^c \geq 0\} - 1 \right] \\
0 &= \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\frac{1}{\sigma}} \frac{(\bar{p}^p + q_{t+1} + d_{t+1})}{p_t^p} \mathbb{I} \{\bar{p}^p - q_{t+1} - d_{t+1} \geq 0\} - 1 \right]
\end{aligned}$$

Question 3 - 4



Using only one moment condition G_1 gives the least accurate estimate. Also, there may also be an identification issue (Two unknown parameters, one moment condition). The OLS estimate of σ is downward biased compared to the true value.